Class 10 Science - Chapter 5

Life Processes (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)
Class 6-7-8 ki yeh cheezein pehle pakki karo, warna Ch5 samajh nahi aayega.

1. Cell - jeevan ki ikai (Class 8 base)

Cell = sabse chhoti structural+functional ikai jeevan ki.
Unicellular = ek cell ka jeev (Amoeba, Paramecium). Multicellular = bahut cell.
Cell -> Tissue -> Organ -> Organ system -> Organism (badhta kram).
Diffusion = padarth ka zyada concentration se kam ki taraf khud jaana.

2. Nutrition basics (Class 7)

Nutrition = bhojan lena aur use energy/growth ke liye use karna.
Autotroph = khud bhojan banaye (green plants - photosynthesis).
Heterotroph = doosron par nirbhar (jaanwar, fungi).
Photosynthesis = sooraj ki roshni me CO_2 + paani se glucose banana.

3. Respiration ka idea (Class 7)

Respiration = bhojan (glucose) ko todkar energy nikalna.
Energy "ATP" form me store hoti (cell ki currency).
Breathing (saans) alag hai - woh sirf hawa andar-bahar; respiration cell ke andar energy banana hai.

4. Body fluids + organs (Class 7-8)

Blood = laal taral jo poore body me oxygen+bhojan le jaata.
Heart = pump jo blood circulate karta. Lungs = saans ke liye.
Stomach/intestine = digestion. Kidney = chhanne (filter) ka kaam.

5. Enzyme aur Gland

Enzyme = biological catalyst (reaction ko fast karta), jaise digestion me.
Gland = woh ang jo ras (juice)/hormone banata (jaise saliva gland).
Oxygen ki kami me anaerobic (bina O_2) respiration ho sakti.

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT
Class 10 Science | Ch 5: Life Processes

[GREEN = board exam me baar-baar poocha jaata hai]

1. Life Processes kya hain?

Life processes = woh kaam jo jeev ko zinda rakhte hain.
4 main: NUTRITION, RESPIRATION, TRANSPORTATION, EXCRETION.
Living ki pehchaan: movement zaroori nahi (molecular level pe motion hoti).

2. NUTRITION [TOP EXAM AREA]

Autotrophic nutrition (green plants): PHOTOSYNTHESIS.
$6CO_2 + 6H_2O$ --(sunlight, chlorophyll)--> $C_6H_{12}O_6 + 6O_2$
Site: chloroplast (chlorophyll = green pigment, light pakadta).
3 raw material: CO_2 (stomata se), water (jad se), sunlight.
Photosynthesis ke steps:
(1) Chlorophyll dwara light energy absorb.
(2) Light energy se water tootta (H aur O) + ATP bante.
(3) CO_2 ka reduction -> glucose (carbohydrate) banta.
STOMATA = patti ke chhote chhidra - gas exchange + transpiration.
Guard cells stomata ko khol/band karte (paani ke hisaab se).

Heterotrophic nutrition ke types:
- Saprophytic: mre/sadhe padarth se (fungi, bacteria) - bahar digest.
- Parasitic: doosre jeev se (Cuscuta, kide, ju) - host ko nuksaan.
- Holozoic: thos bhojan andar lekar (Amoeba, human).

Amoeba me nutrition: pseudopodia se bhojan gher kar food vacuole me digest (phagocytosis). Steps: Ingestion, Digestion, Absorption, Assimilation, Egestion.

3. Human Digestive System [VERY IMPORTANT]

Alimentary canal: Mouth -> Oesophagus (food pipe) -> Stomach -> Small intestine -> Large intestine -> Anus.
Mouth: Saliva (salivary gland) me SALIVARY AMYLASE/ptyalin -> starch ko maltose (sugar) me todta. Daant chabate, jeebh mix karti.
Stomach: Gastric glands se -> (a) HCl (kills bacteria + medium acidic banata), (b) PEPSIN (protein digest karta), (c) mucus (deewar ko HCl se bachata).
Small intestine: sabse zyada digestion + ABSORPTION yahin.
- Liver se BILE (pitt ras): fat ko chhote drops me todta (emulsification), acidic food ko alkaline banata.
- Pancreas se: trypsin (protein), lipase (fat), amylase (starch).
- VILLI (ungli jaise ubhaar) absorption ke liye surface area badhate.

Large intestine: paani absorb, baaki waste anus se bahar (egestion).

4. RESPIRATION [VERY IMPORTANT]

Aerobic respiration (O_2 ke saath, mitochondria me): zyada energy.
Glucose --> CO_2 + H_2O + ENERGY (bahut ATP).
Anaerobic (bina O_2):
- Yeast me: Glucose -> Ethanol + CO_2 + energy (fermentation).
- Hamari muscles me (zyada exercise, O_2 kam): Glucose -> LACTIC ACID + energy
  -> isi se muscle me CRAMP/dard hota.

Glucose pehle CYTOPLASM me PYRUVATE me tootta (3-carbon), phir:
O_2 ho -> mitochondria me CO_2+H_2O ; O_2 na ho -> lactic acid/ethanol.
Mitochondria = "power house of cell" (ATP banta).

Human Respiratory System:
Nostrils -> Pharynx -> Larynx -> Trachea (windpipe) -> Bronchi ->
Bronchioles -> ALVEOLI (lungs me).
ALVEOLI = thaili jaise, gas exchange yahin (O_2 blood me, CO_2 bahar).
Surface area bahut zyada (balloon jaise) + patli deewar + bahut blood vessel.
Trachea me cartilage ke ring -> hawa ka raasta band nahi hota.
Haemoglobin (RBC me) -> O_2 ko jodke poore body me le jaata.
CO_2 paani me zyada ghulta -> isliye blood ke through (na ki hawa) jaata.

5. TRANSPORTATION (Human) [TOP EXAM AREA]

Heart: 4 chambers - 2 Atria (upar) + 2 Ventricles (neeche).
Right side = de-oxygenated (CO_2 wala) blood; Left side = oxygenated.
Double circulation: blood 2 baar heart se guzarta (pulmonary + systemic).
Blood ka rasta:
Body -> Right atrium -> Right ventricle -> Lungs (oxygen mila) ->
Left atrium -> Left ventricle -> Body (poore body ko O_2).
Left ventricle ki deewar SABSE MOTI (poore body me blood pump karna padta).
VALVES (kapaat): blood ko peeche jaane se rokte (one-way).

Blood vessels (3 type):
- ARTERY: heart se body ki taraf, oxygenated (except pulmonary artery),
  moti deewar, koi valve nahi, blood high pressure me.
- VEIN: body se heart ki taraf, de-oxygenated (except pulmonary vein),
  patli deewar, valves hote (backflow rokne).
- CAPILLARY: sabse patli (ek cell moti) - yahin exchange hota.

Blood components: Plasma (taral), RBC (O_2-haemoglobin), WBC (bachav),
Platelets (clotting/khoon jamna).
Lymph = ek aur fluid, fat absorb + body defence (RBC nahi hota).

Plants me transportation:
- XYLEM: paani + minerals ko JAD se PATTI tak (upar) le jaata.
  TRANSPIRATION PULL (patti se paani udta -> kheechta) main force.
- PHLOEM: bana hua bhojan (glucose) ko patti se baaki body tak
  (TRANSLOCATION) - dono dishaon me, ATP energy use karke.

6. EXCRETION [IMPORTANT]

Excretion = body se nitrogen-waste (jaise urea) nikalna.

Human excretory system: 2 Kidney + Ureter + Urinary bladder + Urethra.
NEPHRON = kidney ki functional ikai (filter unit). Yahin blood filter hota.

- Glomerulus me blood filter; useful cheez (glucose, salt, paani) WAPAS absorb (reabsorption); waste urine ban jaata.

DIALYSIS = jab kidney fail ho to machine se artificial blood filter.
Plants me excretion: O_2/CO_2 stomata se, extra paani transpiration se, kuch waste patti girakar ya gum/resin ke roop me.

7. Quick Revision - One Liners

- Photosynthesis: $6CO_2+6H_2O$--sunlight/chlorophyll-->$C_6H_{12}O_6+6O_2$.
- Aerobic (mito, zyada energy); anaerobic (lactic acid - cramp / ethanol-yeast).
- Heart 4 chambers; left ventricle moti deewar; double circulation.
- Artery (heart se, no valve); Vein (heart ko, valves); Capillary (exchange).
- Xylem paani upar; Phloem bhojan; Nephron kidney ki ikai.

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Double circulation poora samjhao

Q: Manav heart me "double circulation" kya hai? Blood ka pura rasta + left ventricle ki deewar moti kyun?

Solution:
- Double circulation: ek complete chakkar me blood DO baar heart se guzarta. (1) Pulmonary: heart -> lungs -> heart (O_2 lene). (2) Systemic: heart -> body -> heart (O_2 dene).
- Rasta: Body(CO_2) -> Right atrium -> Right ventricle -> Lungs -> Left atrium -> Left ventricle -> poore Body.
- Left ventricle ko poore sharir me blood high pressure se bhejna padta, isliye uski deewar sabse moti+strong.
- Faayda: oxygenated aur de-oxygenated blood mix nahi hote -> efficient.

Example 2 - Photosynthesis steps + raw materials

Q: Photosynthesis ka equation, site, raw materials aur 3 steps likho.

Solution:
- Equation: $6CO_2 + 6H_2O$ --sunlight/chlorophyll--> $C_6H_{12}O_6 + 6O_2$.
- Site: chloroplast (chlorophyll pigment).
- Raw material: CO_2 (stomata), paani (jad), sunlight.
- Steps: (1) light absorb, (2) water tootna + ATP, (3) CO_2 reduce -> glucose.

Example 3 - Anaerobic respiration aur cramp

Q: Zyada daudne par muscles me dard (cramp) kyun? Reaction likho.

Solution:
- Zyada exercise me O_2 kam pad jaata -> muscle anaerobic respiration karta.
- Glucose -> Lactic acid + energy. Lactic acid jamne se cramp/dard.
- Aaram + O_2 milne par lactic acid khatam, dard kam.

Example 4 - Artery vs Vein

Q: Artery aur vein me 3 antar likho.

Solution:
- Artery: heart se bahar; oxygenated (except pulmonary); moti deewar; no valve.
- Vein: heart ki taraf; de-oxygenated (except pulmonary); patli deewar; valves.

Example 5 (EASIEST) - Life processes ginvao

Q: 4 main life processes kaunse hain?

Solution:
Nutrition, Respiration, Transportation, Excretion.

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

Q1. Autotroph aur heterotroph me antar.
Q2. Stomata ka kaam kya hai?
Q3. Mitochondria ko cell ka kya kehte aur kyun?
Q4. Xylem aur phloem ka kaam ek-ek line.

MEDIUM (2-3 marks)

Q5. Photosynthesis ka equation + raw materials + site.
Q6. Aerobic aur anaerobic respiration me antar (product sahit).
Q7. Human digestion me HCl, pepsin, bile ka kaam.
Q8. Nephron kya hai? Kidney me filter + reabsorption samjhao.

HARD (3-5 marks, board favourite)

Q9. Double circulation + blood ka pura rasta + left ventricle moti kyun.
Q10. Human respiratory system ka path + alveoli ki khaasiyat.
Q11. Small intestine me digestion (liver, pancreas, villi ka role).
Q12. Plants me water transport - xylem + transpiration pull samjhao.

ANSWER HINTS (PART D)

A1. Autotroph khud bhojan banaye (plant); heterotroph doosron par nirbhar.
A2. Gas exchange + transpiration (paani vapor nikalna).
A3. "Power house" - kyunki yahin ATP (energy) banta.
A4. Xylem paani+mineral upar; phloem bhojan poore plant me.
A5. $6CO_2+6H_2O$--sun/chlorophyll-->$C_6H_{12}O_6+6O_2$; CO_2/water/sunlight; chloroplast.
A6. Aerobic O_2 ke saath mito me CO_2+H_2O+jyada energy; anaerobic bina O_2, lactic acid (muscle) ya ethanol+CO_2 (yeast).
A7. HCl bacteria maare+acidic; pepsin protein; bile fat emulsify+alkaline.
A8. Nephron = kidney ki filter ikai; glomerulus filter, useful cheez reabsorb, waste urine.
A9. Blood 2 baar heart se; body->RA->RV->lungs->LA->LV->body; LV moti (high pressure pump).
A10. Nostril->trachea->bronchi->bronchioles->alveoli; alveoli surface area zyada+patli deewar+blood vessel.
A11. Bile (fat emulsify), pancreatic enzyme (trypsin/lipase/amylase), villi absorption surface badhate.
A12. Xylem jad se patti; transpiration pull (patti se paani udke kheechta).

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!
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